

Handling marine movement data

Marie Auger-Méthé

Canadian Research Chair Tier II in Statistical Ecology

Institute for the Oceans & Fisheries

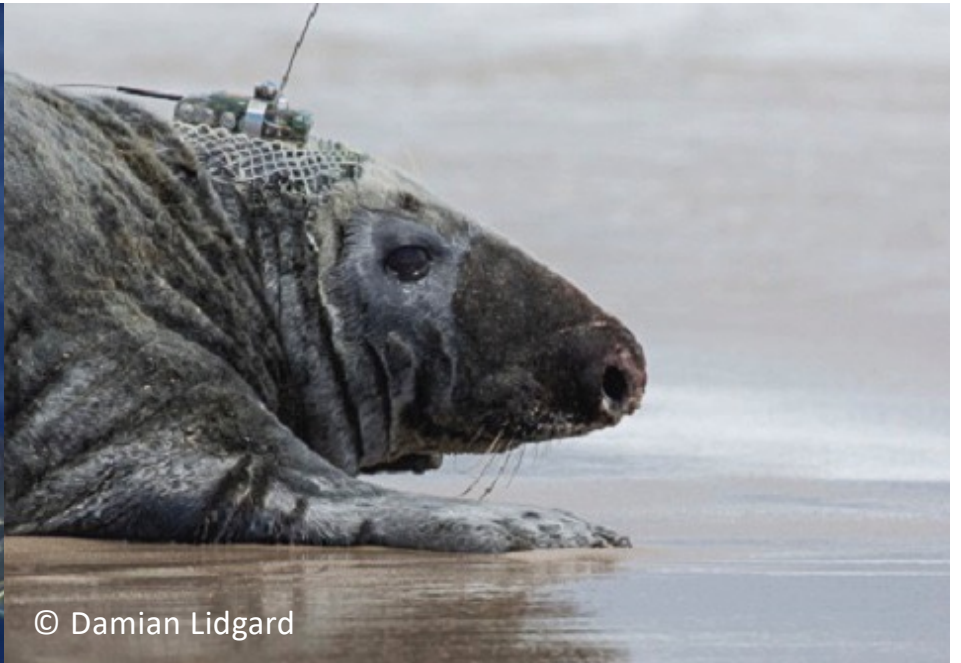
Dept. of Statistics

University of British Columbia



Problems with marine tracking

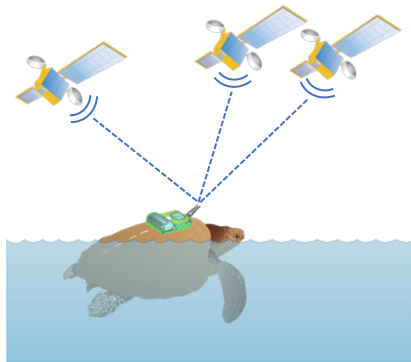
- GPS doesn't work underwater
- Animals don't surface long enough to get to GPS location



Alternatives for animals that surface

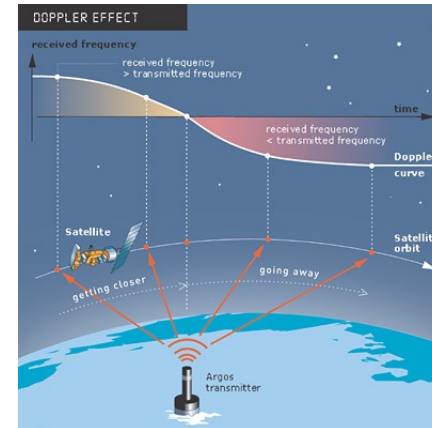
Fastloc GPS

More accurate
More expensive

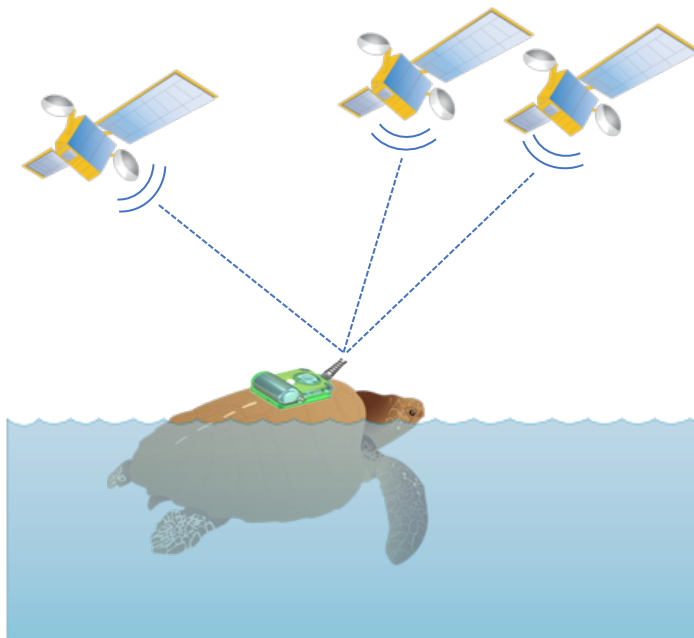


Argos

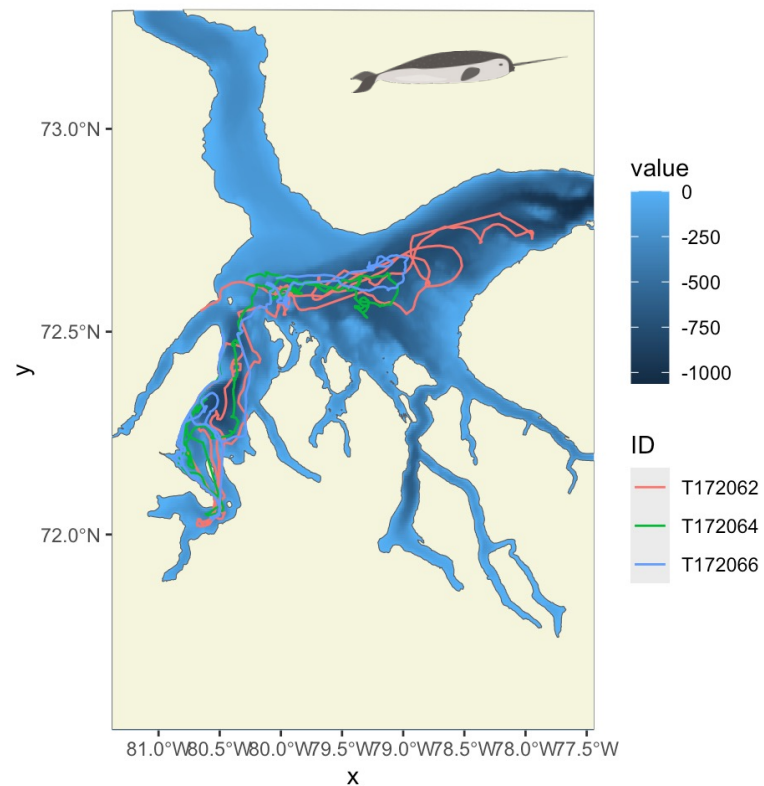
Less accurate
Less expensive



Fastloc GPS

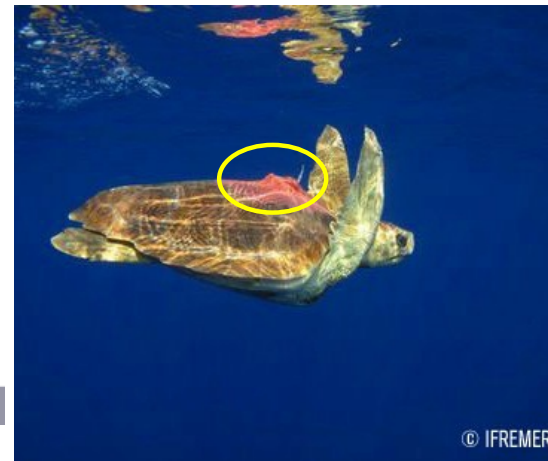


Fastloc GPS

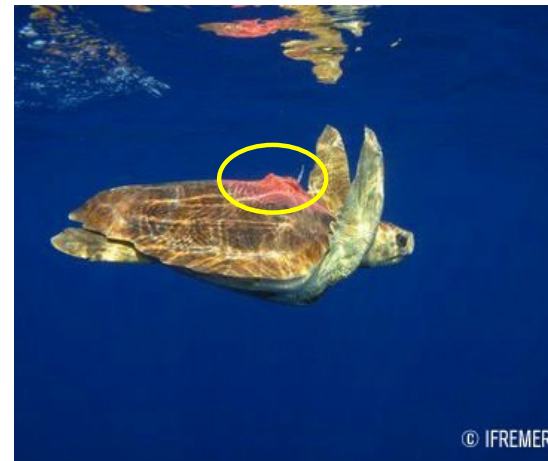
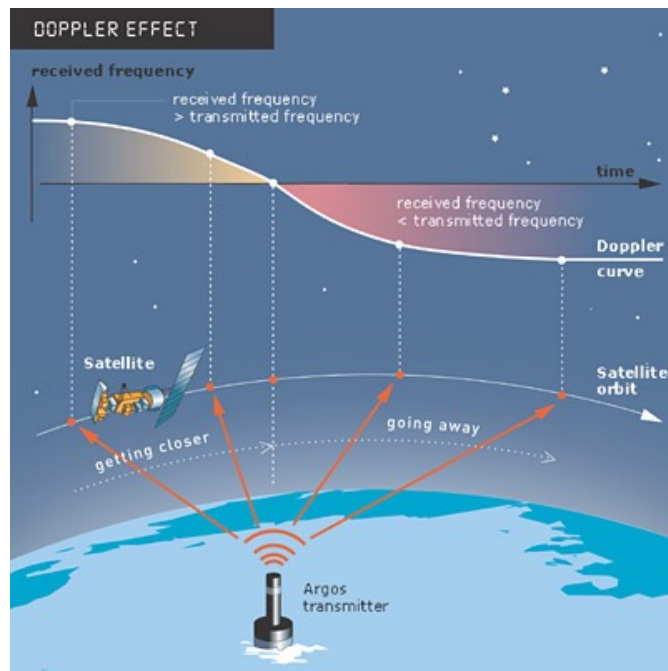


- Can be used on animals that surface for only 10s of milliseconds
- Accurate <200 m

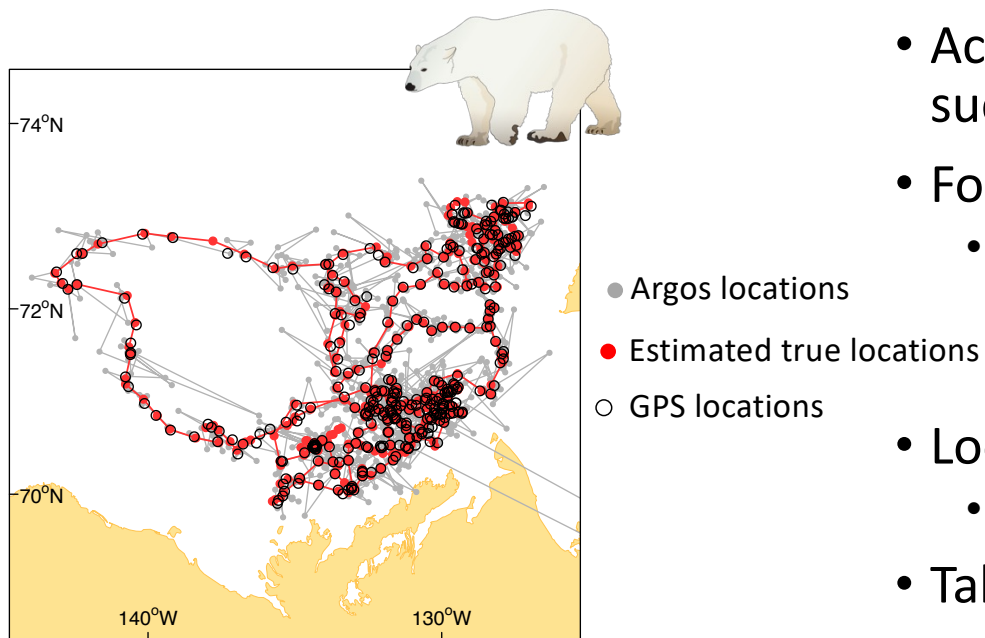
Argos



Argos

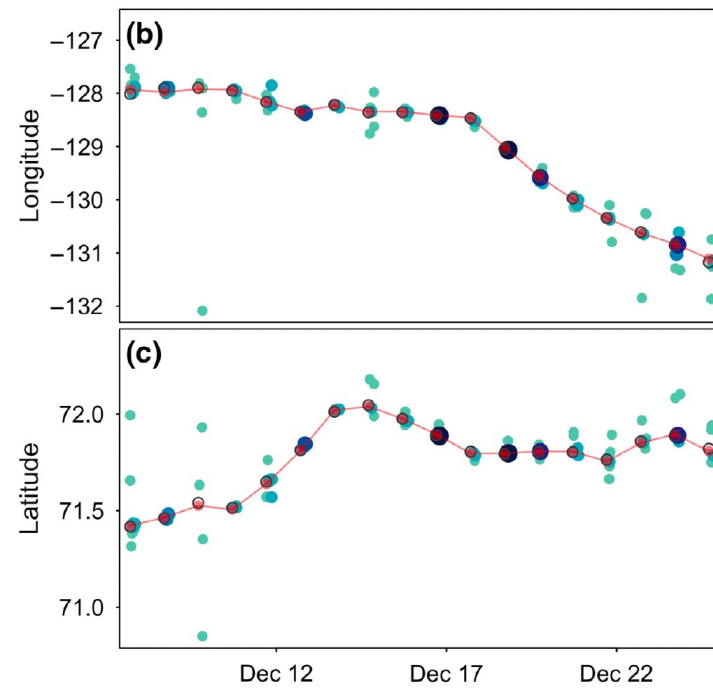
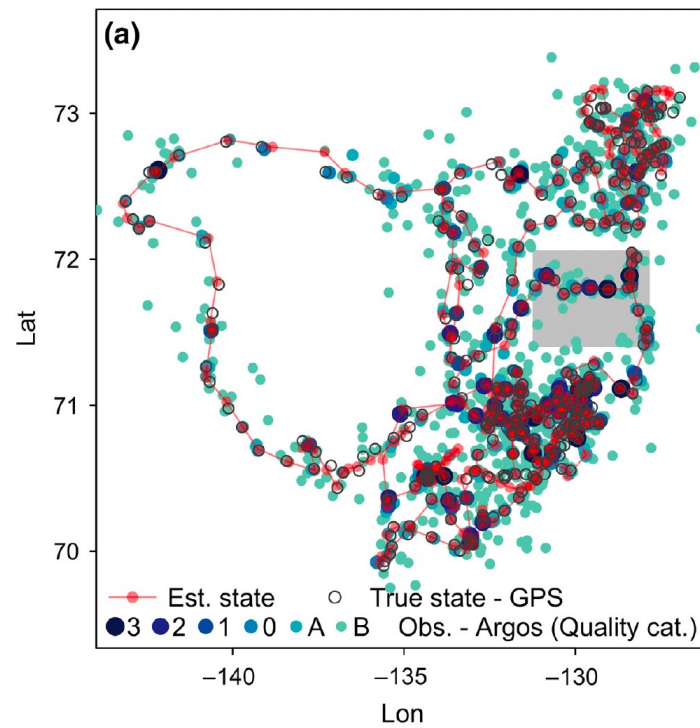


Argos



- Accuracy relies on the number of successive satellite uplinks
- For marine species, low accuracy
 - error of 4–36 km
- Location quality:
 - 3, 2, 1 0, A, B, Z
- Taken at irregular time

Argos

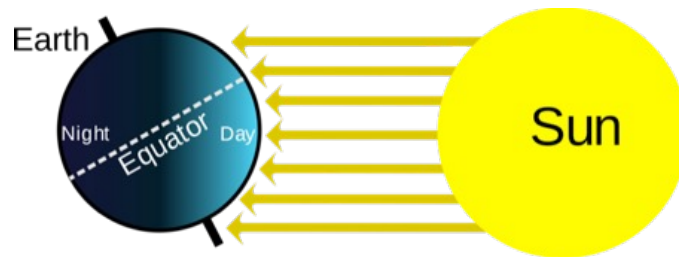


Auger-Méthé et al. (2021) Ecological Monographs

Alternative for animals that stay under (or small animals)

Light-based geolocation

Latitudes use daylength



Longitudes use time difference
between local and Greenwich time



Light-based geolocation

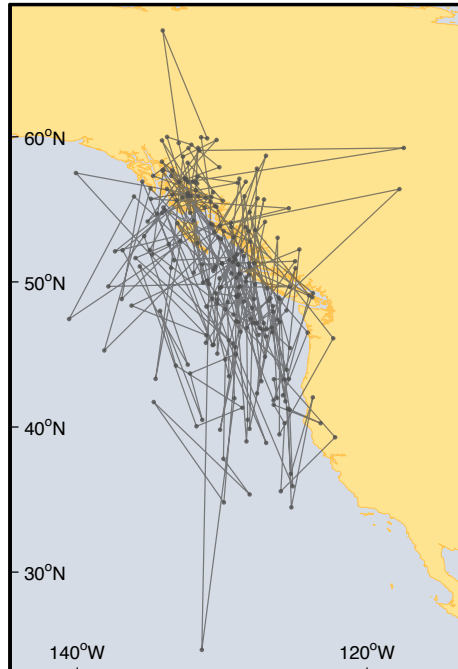


Photos: Katharine Studholme, M. Drever

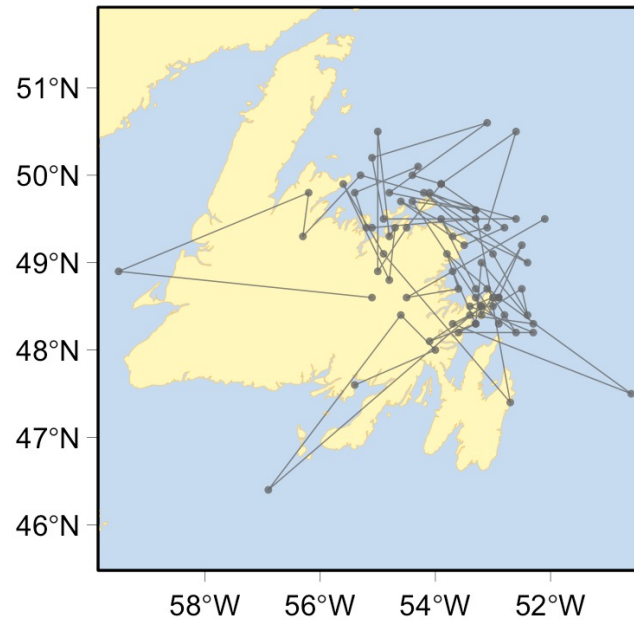
Light-based geolocation



Rhinoceros auklet

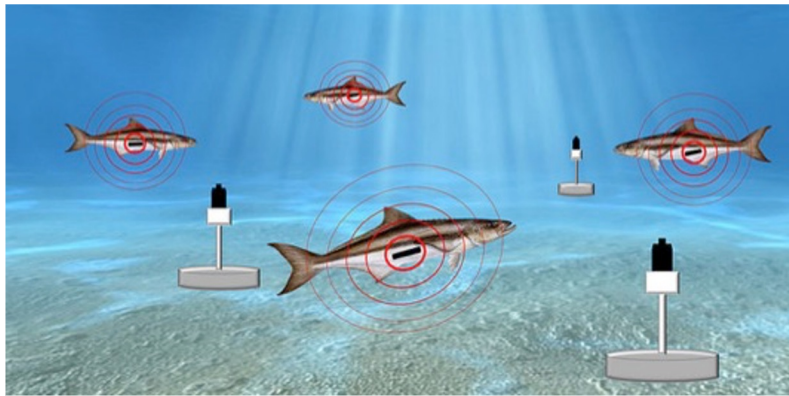


Atlantic salmon



- Very inaccurate data
 - Error 200km

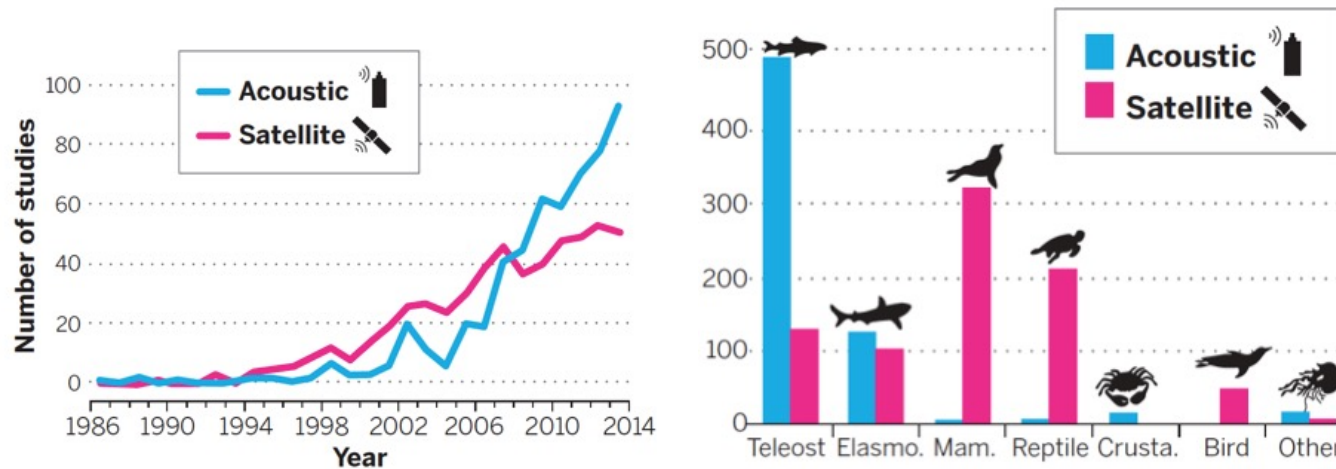
Acoustic tracking



F.W.C. 2016



Increase in telemetry studies



Aquatic animal telemetry: A panoramic window into the underwater world

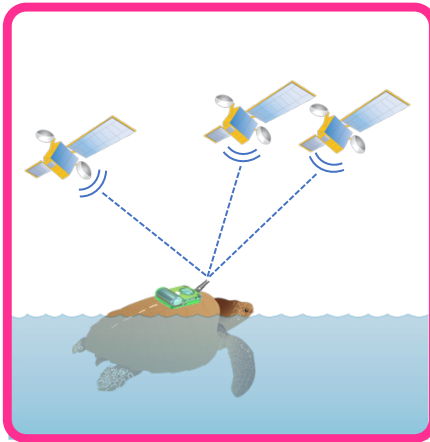


Nigel E. Hussey,¹ Steven T. Kessel,¹ Kim Aarestrup,² Steven J. Cooke,³ Paul D. Cowley,⁴
Aaron T. Fisk,¹ Robert G. Harcourt,⁵ Kim N. Holland,⁶ Sara J. Iverson,^{7*} John F. Kocik,⁸
Joanna E. Mills Flemming,⁹ Fred G. Whoriskey⁷

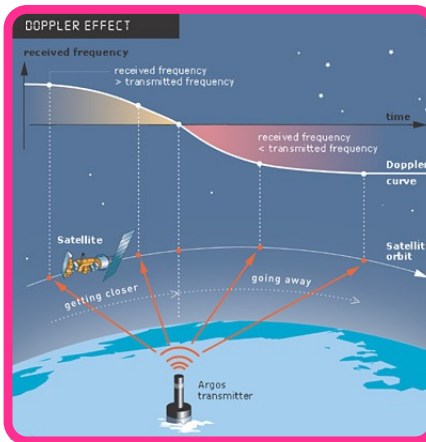
Comparison between methods

Irregular in time

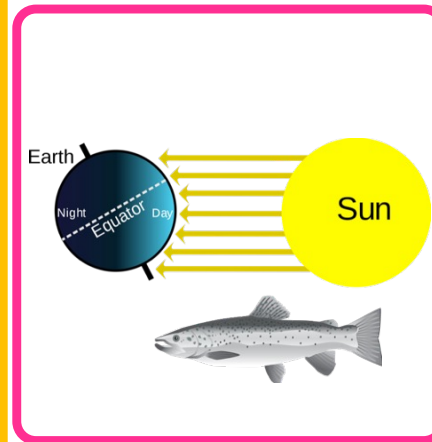
Fastloc GPS



Argos



Light-based
geolocation



Low



Error

High

How do we handle error and
irregular time steps?

Irregular time steps

- Continuous time models
- Split tracks with data gap
- Regularize the time series
 - Simple interpolate
 - Prediction from a more complex model

State-space model to handle error

Process equation:

$$x_t = f(x_{t-1}) + \eta_t$$

State Behavioural mechanism Process noise

Measurement equation:

$$y_t = g(x_t) + \epsilon_t$$

Data Observation mechanism Measurement error

State-space model to handle error

Process equation:

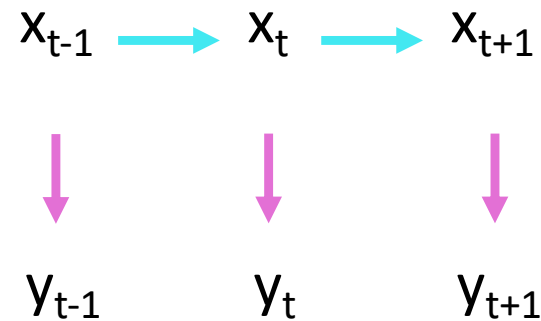
$$x_t = f(x_{t-1}) + \eta_t$$

State Behavioural mechanism Process noise

Measurement equation:

$$y_t = g(x_t) + \epsilon_t$$

Data Observation mechanism Measurement error



State-space model to handle error

Process equation:

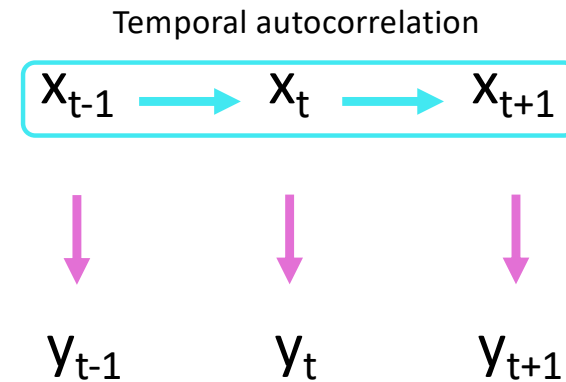
$$x_t = f(x_{t-1}) + \eta_t$$

State Behavioural mechanism Process noise

Measurement equation:

$$y_t = g(x_t) + \epsilon_t$$

Data Observation mechanism Measurement error



State-space models

Two-step approach (today's focus)

- Step 1: simple model to remove error (and regularize)
- Step 2: analysis of interest
 - (e.g., resource selection function)
- Pros:
 - Many R packages
 - Fast to fit
- Con:
 - don't propagate error (except if we use multiple imputation)

- Model that handles error and includes analysis of interest (Friday's focus)

- Pro:
 - propagate error
- Cons:
 - Often involves hard-coding model
 - Often slow

State-space models

Two-step approach (today's focus)

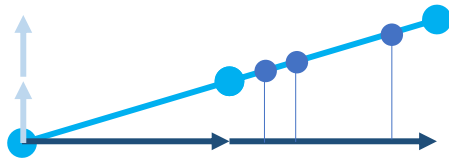
- **Step 1: simple model to remove error (and regularize)**
- Step 2: analysis of interest
 - (e.g., resource selection function)
- Pros:
 - Many R packages
 - Fast to fit
- Con:
 - don't propagate error (except if we use multiple imputation)

- Model that handles error and includes analysis of interest (Friday's focus)

- Pro:
 - propagate error
- Cons:
 - Often involves hard-coding model
 - Often slow

CTCRW with crawl (via momentuHMM)

Process equation (basic concept):



Velocity (speed)

$$v(t + \Delta) = e^{-\beta\Delta} v(t)$$

How quickly the temp.
correlation $\rightarrow 0$

$\Delta \uparrow$
correlation in $v \downarrow$

Location

$$\mu(t + \Delta) = \mu(t) + \int_t^{t+\Delta} v(u) du$$

Ecology, 89(5), 2008, pp. 1208–1215
© 2008 by the Ecological Society of America

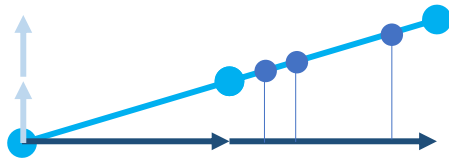
CONTINUOUS-TIME CORRELATED RANDOM WALK MODEL
FOR ANIMAL TELEMETRY DATA

DEVIN S. JOHNSON,¹ JOSHUA M. LONDON, MARY-ANNE LEA,² AND JOHN W. DURBAN

National Marine Mammal Laboratory, Alaska Fisheries Science Center, National Marine Fisheries Service

CTCRW with crawl (via momentuHMM)

Process equation (basic concept):



Velocity (speed)

How quickly the temp. correlation $\rightarrow 0$

$$v(t + \Delta) = e^{-\beta\Delta} v(t) + \zeta(\Delta)$$

$\Delta \uparrow$
correlation in $v \downarrow$

$$N\left(0, \frac{\sigma^2[1 - e^{-2\beta\Delta}]}{2\beta}\right)$$

Location

$$\mu(t + \Delta) = \mu(t) + \int_t^{t+\Delta} v(u) du$$

Ecology, 89(5), 2008, pp. 1208–1215
© 2008 by the Ecological Society of America

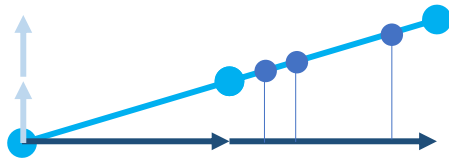
CONTINUOUS-TIME CORRELATED RANDOM WALK MODEL
FOR ANIMAL TELEMTRY DATA

DEVIN S. JOHNSON,¹ JOSHUA M. LONDON, MARY-ANNE LEA,² AND JOHN W. DURBAN

National Marine Mammal Laboratory, Alaska Fisheries Science Center, National Marine Fisheries Service

CTCRW with crawl (via momentuHMM)

Process equation (basic concept):



Velocity (speed)

$$v(t + \Delta) = e^{-\beta\Delta} v(t) + \zeta(\Delta), \quad \zeta(\Delta) \sim N\left(0, \frac{\sigma^2[1 - e^{-2\beta\Delta}]}{2\beta}\right)$$

Location

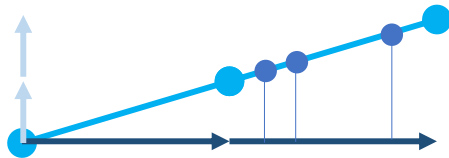
$$\mu(t + \Delta) = \mu(t) + \int_t^{t+\Delta} v(u) du$$

Measurement equation:

$$y(t) = \mu(t) + \epsilon(t), \quad \epsilon(t) \sim N(0, \sigma_m^2)$$

CTCRW with crawl (via momentuHMM)

Process equation (basic concept):



Velocity (speed)

$$v(t + \Delta) = e^{-\beta\Delta}v(t) + \zeta(\Delta), \quad \zeta(\Delta) \sim N\left(0, \frac{\sigma^2[1 - e^{-2\beta\Delta}]}{2\beta}\right)$$

Location

$$\mu(t + \Delta) = \mu(t) + \int_t^{t+\Delta} v(u)du$$

Measurement equation:

$$y(t) = \mu(t) + \epsilon(t), \quad \epsilon(t) \sim N(0, \sigma_m^2)$$

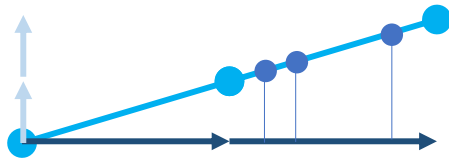
Ecology, 89(5), 2008, pp. 1208–1215
© 2008 by the Ecological Society of America

CONTINUOUS-TIME CORRELATED RANDOM WALK MODEL
FOR ANIMAL TELEMETRY DATA

DEVIN S. JOHNSON,¹ JOSHUA M. LONDON, MARY-ANNE LEA,² AND JOHN W. DURBAN

CTCRW with crawl (via momentuHMM)

Process equation (basic concept):



Velocity (speed)

$$v(t + \Delta) = e^{-\beta \Delta} v(t) + \zeta(\Delta), \quad \zeta(\Delta) \sim N\left(0, \frac{\sigma^2 [1 - e^{-2\beta \Delta}]}{2\beta}\right)$$

Location

$$\mu(t + \Delta) = \mu(t) + \int_t^{t+\Delta} v(u) du$$

Measurement equation:

$$y(t) = \mu(t) + \epsilon(t), \quad \epsilon(t) \sim N(0, \sigma_m^2)$$

Ecology, 89(5), 2008, pp. 1208–1215
© 2008 by the Ecological Society of America

CONTINUOUS-TIME CORRELATED RANDOM WALK MODEL
FOR ANIMAL TELEMETRY DATA

DEVIN S. JOHNSON,¹ JOSHUA M. LONDON, MARY-ANNE LEA,² AND JOHN W. DURBAN

aniMotum – CRW version

Process equation:

Velocity (speed)

$$v(t + \Delta) = v(t) + \zeta(\Delta), \quad \zeta(\Delta) \sim N(0, 2D\Delta)$$

Location

$$\mu(t + \Delta) = \mu(t) + v(t + \Delta)\Delta$$

Measurement equation:

$$y(t) = \mu(t) + \epsilon(t), \epsilon(t) \sim N(0, \Sigma)$$

Jonsen et al. *Movement Ecology* (2020) 8:31
<https://doi.org/10.1186/s40462-020-00217-7>

Movement Ecology

METHODOLOGY ARTICLE

Open Access

A continuous-time state-space model for rapid quality control of argos locations from animal-borne tags

Ian D. Jonsen^{1*}, Toby A. Patterson², Daniel P. Costa³, Philip D. Doherty⁴, Brendan J. Godley⁴, W. James



aniMotum – mp version

Process equation:

Velocity (speed)

$$v(t + \Delta) = \gamma(t + \Delta)v(t)$$

Location

$$\mu(t + \Delta) = \mu(t) + v(t + \Delta)\Delta + \xi(\Delta), \xi(\Delta) \sim N(0, \Delta^2 \sigma^2)$$

Measurement equation:

$$y(t) = \mu(t) + \epsilon(t), \epsilon(t) \sim N(0, \Sigma)$$

Ecology, 100(1), 2019, e02566
© 2018 by the Ecological Society of America

Movement responses to environment: fast inference of variation
among southern elephant seals with a mixed effects model

I. D. JONSEN ^{1,6} C. R. McMAHON ² T. A. PATTERSON,³ M. AUGER-MÉTHÉ,⁴ R. HARCOURT,¹
M. A. HINDELL,⁵ AND S. BESTLEY⁵

Vol. 565: 237–249, 2017
doi: 10.3354/meps12019

MARINE ECOLOGY PROGRESS SERIES
Mar Ecol Prog Ser

Published February 17

Spatiotemporal modelling of marine movement data using Template Model Builder (TMB)

Marie Auger-Méthé^{1,*}, Christoffer M. Albertsen², Ian D. Jonsen³,
Andrew E. Derocher⁴, Damian C. Lidgard⁵, Katharine R. Studholme⁵,

aniMotum – mp version

Process equation:

$$\text{logit}(\gamma(t + \Delta)) = \text{logit}(\gamma(t)) + \varsigma(\Delta), \quad \varsigma(\Delta) \sim N(0, \Delta\sigma_\gamma^2)$$

Velocity (speed)

$$v(t + \Delta) = \gamma(t + \Delta)v(t)$$

Location

$$\mu(t + \Delta) = \mu(t) + v(t + \Delta)\Delta + \xi(\Delta), \quad \xi(\Delta) \sim N(0, \Delta^2\sigma^2)$$

Measurement equation:

$$y(t) = \mu(t) + \epsilon(t), \quad \epsilon(t) \sim N(0, \Sigma)$$

Ecology, 100(1), 2019, e02566
© 2018 by the Ecological Society of America

Movement responses to environment: fast inference of variation
among southern elephant seals with a mixed effects model

I. D. JONSEN ^{1,6} C. R. McMAHON ² T. A. PATTERSON,³ M. AUGER-MÉTHÉ,⁴ R. HARCOURT,¹
M. A. HINDELL,⁵ AND S. BESTLEY⁵

Vol. 565: 237–249, 2017
doi: 10.3354/meps12019

MARINE ECOLOGY PROGRESS SERIES
Mar Ecol Prog Ser

Published February 17

**Spatiotemporal modelling of marine movement
data using Template Model Builder (TMB)**

Marie Auger-Méthé^{1,*}, Christoffer M. Albertsen², Ian D. Jonsen³,
Andrew E. Derocher⁴, Damian C. Lidgard⁵, Katharine R. Studholme⁵,

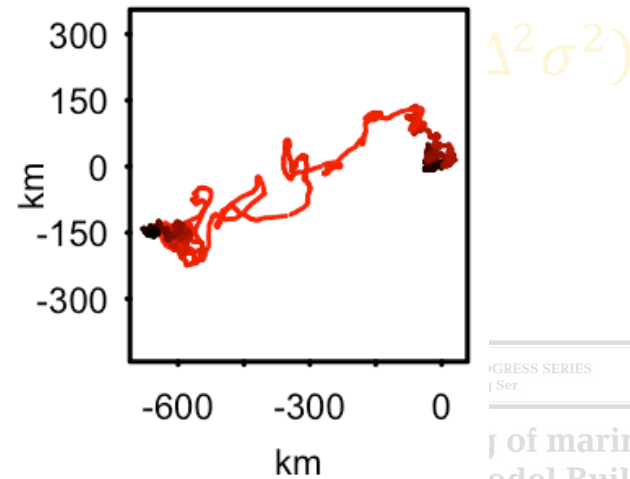
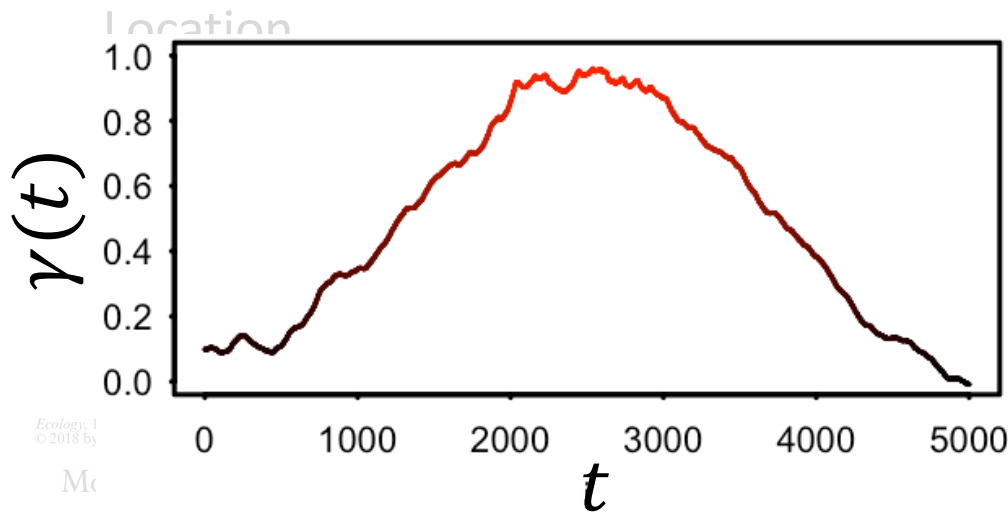
aniMotum – mp version

Process equation:

Velocity (speed)

$$v(t + \Delta) = \gamma(t + \Delta)v(t)$$

$$\text{logit}(\gamma(t + \Delta)) = \text{logit}(\gamma(t)) + \varsigma(\Delta), \quad \varsigma(\Delta) \sim N(0, \Delta\sigma_\gamma^2)$$



Ecology, 1
© 2018 by

Mc

I. D. JONSEN^{1,6}, C. R. MCMAHON², T. A. PATTERSON³, M. AUGER-MÉTHÉ⁴, R. HARCOURT¹,
M. A. HINDELL⁵ AND S. BESTLEY⁵

GRESS SERIES
f Ser

Published February 17

Journal of marine movement
Model Builder (TMB)

Marie Auger-Méthé^{1,*}, Christoffer M. Albertsen², Ian D. Jonsen³,
Andrew E. Derocher⁴, Damian C. Lidgard⁵, Katharine R. Studholme⁵,

aniMotum – mp version

Process equation:

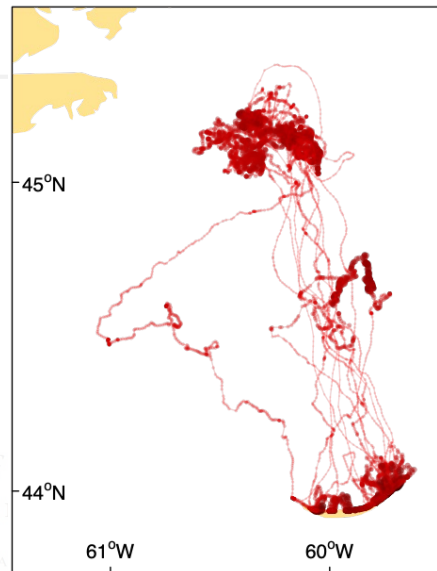
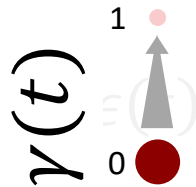
Velocity (speed)

$$v(t + \Delta) = \gamma(t + \Delta)v(t)$$

Location

$$\mu(t + \Delta) = \mu(t) + v(t + \Delta)\Delta$$

Discrete
approximation:
small Δ



© Damian Lidgard

Ecology
© 2018 by the author(s)

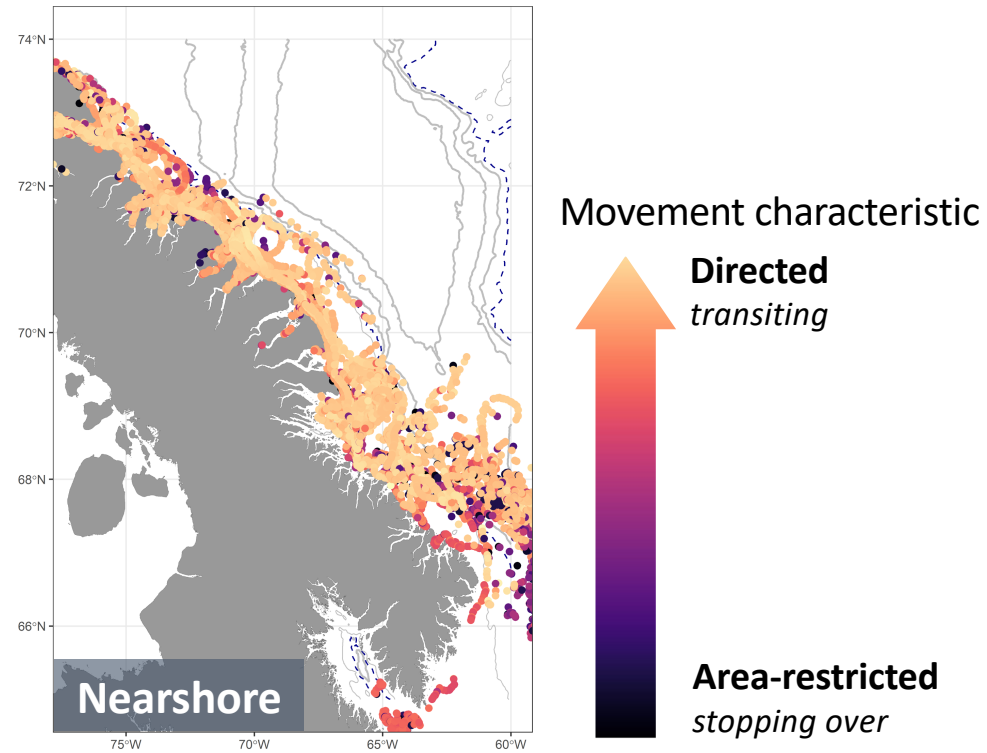
Movement responses to environment: fast inference of
among southern elephant seals with a mixed effects

I. D. JONSEN^{1,6}, C. R. McMAHON², T. A. PATTERSON³, M. AUGER-MÉTÉ⁴, R. H.
M. A. HINDELL⁵ AND S. BESTLEY⁵

Auger-Méthé^{1,*}, Christoffer M. Albertsen², Ian D. Jonsen³,
Andrew E. Derocher⁴, Damian C. Lidgard⁵, Katharine R. Studholme⁵,



Migration corridors and stop-over sites

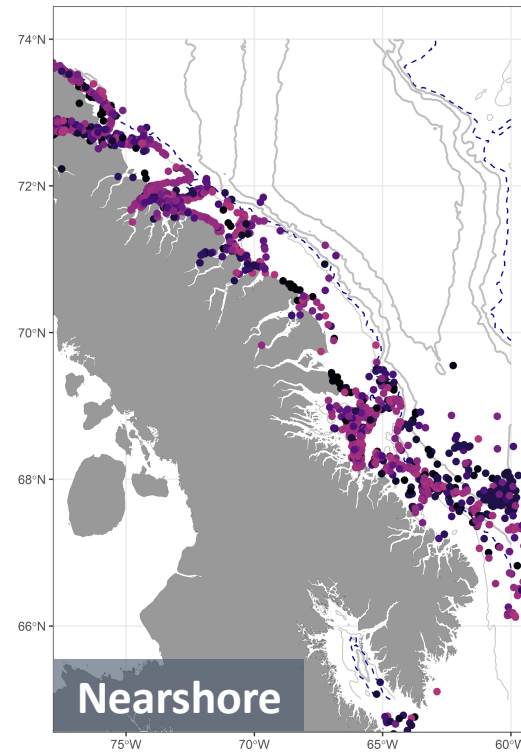


Project collaborators: Marianne Marcoux, Nigel Hussey, ...

Shuert et al. 2023 Movement Ecology



Migration corridors and stop-over sites



Movement characteristic

Directed
transiting

Area-restricted
stopping over



Courtney Shuert
Former postdoc

Project collaborators: Marianne Marcoux, Nigel Hussey, ...

Shuert et al. 2023 Movement Ecology

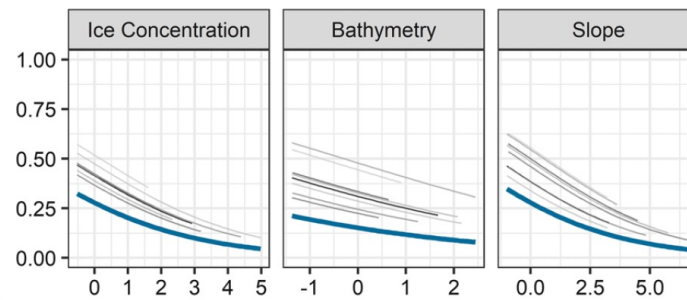
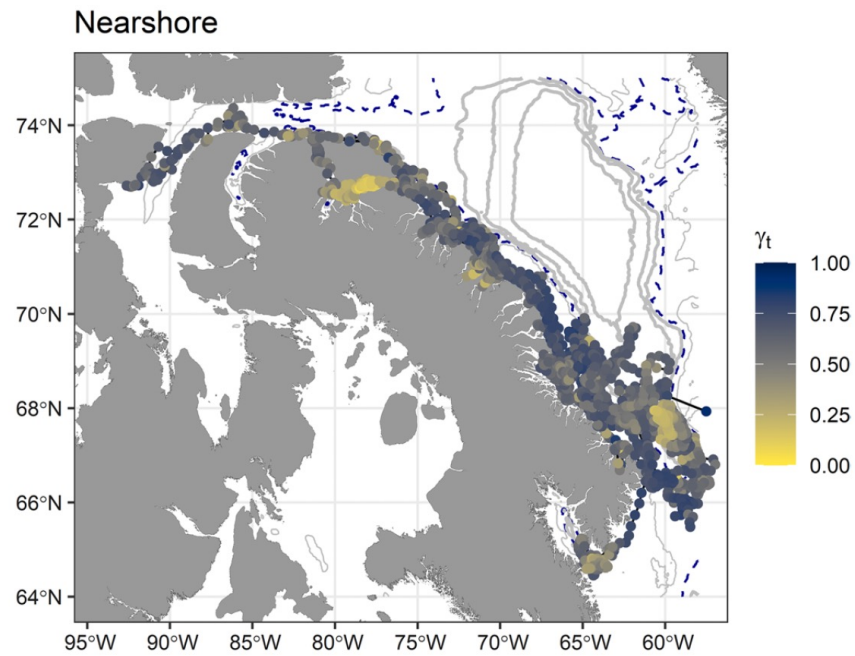
State-space models

Two-step approach (today's focus)

- Step 1: simple model to remove error (and regularize)
- **Step 2: analysis of interest**
 - (e.g., resource selection function)
- Pros:
 - Many R packages
 - Fast to fit
- Con:
 - don't propagate error (except if we use multiple imputation)

- Model that handles error and includes analysis of interest (Friday's focus)

- Pro:
 - propagate error
- Cons:
 - Often involves hard-coding model
 - Often slow



Project collaborators: Marianne Marcoux, Nigel Hussey, ...

Shuert et al. 2023 Movement Ecology

State-space models

Two-step approach (today's focus)

- Step 1: simple model to remove error (and regularize)
- **Step 2: analysis of interest**
 - (e.g., resource selection function)
- Pros:
 - Many R packages
 - Fast to fit
- Con:
 - don't propagate error (except if we use multiple imputation)

- Model that handles error and includes analysis of interest (Friday's focus)

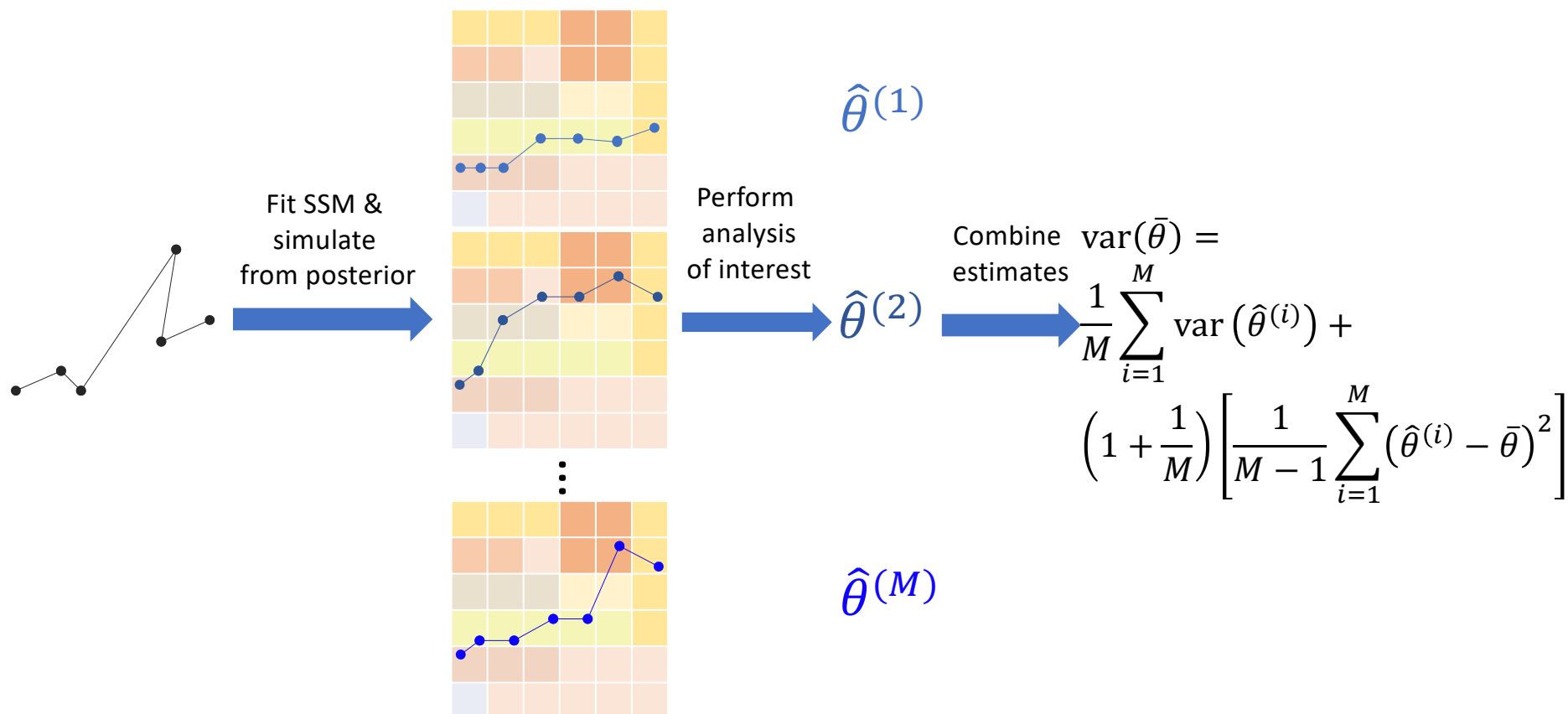
- Pro:
 - propagate error
- Cons:
 - Often involves hard-coding model
 - Often slow

Multiple imputation

**Incorporating Telemetry Error into Hidden
Markov Models of Animal Movement Using
Multiple Imputation**

Brett T. MCCLINTOCK 

Multiple imputation



Let's start applying these
concepts in R!